ICV1 IDENTITIES • DISSECT TRAITS • VERIFIED FOLLOW-UP

Elliptic-curve assessment

Four supplied candidates over the NIST P-224 and P-256 prime fields. This report combines the original computation and the completed follow-up. The candidates have their own coefficients and are distinct from the standardized NIST curves.

Default trait calls	Assessed records with all fields	Output corrections
292 across 22 traits	290 / 292	10 records; 2 source defects

Principal findings

All four supplied curves have proven prime full group order, cofactor **h = 1**, and a generator satisfying **nG = O**. The seeds, generator-selection rule and root conventions match. Each curve is ordinary and non-anomalous, with certified endomorphism conductor **1**.

Input candidate	Curve work ≈ bits	Largest twist prime: bits	Twist work ≈ bits
224 / 777	112.0	83	41.5
224 / 1485	112.0	122	60.5
224 / 5736	112.0	106	52.7
256 / 876	128.0	201	100.2

Work estimates are approximate $\log_2$ group-operation counts from the square root of the relevant prime order. They are mathematical generic-DLP estimates, rather than measured attacks or a security certification. Twist figures concern twist points; proper point validation keeps protocol operations in the supplied curve subgroup.

What is complete, and what remains

The follow-up closed **three of five** missing auxiliary results. Two quadratic-twist factorizations over Fp^2 remain partial, with proven-composite residuals of **327 bits** and **444 bits**. All base-field curve and twist orders are fully certified.

The original 292-call DiSSECT run is preserved. The 290/292 figure describes a labeled assessed view that incorporates verified corrections and recovered factorizations; it does not claim a repeat of every default call.

Evidence included

The complete evidence ZIP, ICV1 JSON and the two-line source patch are embedded as PDF attachments. The ZIP includes raw outputs, the assessed dataset, certificates, reproduction scripts, source snapshots and hash manifests. Some browser PDF viewers require downloading the PDF to access attachments.

Candidate numbers are preserved as supplied labels. The search iterations, rejection history and any statistical outlier comparison against a simulated population were not reconstructed.

EXECUTION AND VERIFICATION

Methods and DiSSECT corrections

Independent arithmetic certificates

Full group order. Proof-enabled SageMath establishes primality of p and n. Independent affine arithmetic and SageMath both verify that the supplied nonidentity G is killed by n. Lagrange's theorem and the unique multiple of n in the Hasse interval certify the full cardinality and h = 1.

Endomorphism ring and embedding degree. Every Frobenius discriminant $D = t^2 - 4p$ is factored into proven primes and is fundamental. For an ordinary curve this gives $\mathbb{Z}[\pi] = \mathrm{End}(E) = \mathcal{O}_K$. Embedding degrees are checked by modular exponentiation and prime-divisor minimality tests.

Seed and root choices. The X9.62 SHA-1 expansion produces c with $b^2c = a^3 \bmod p$. Each 224-bit b is the smaller root; the 256-bit b equals $(a^3/c)^{(p+1)/4} \bmod p$, including its sign. For the 224-bit field, changing b to −b is an Fp-isomorphism. For the 256-bit field it gives the quadratic twist.

Verified source defects

Torsion extension. In the repeated-eigenvalue branch, the full point-field degree must be multiplied by ℓ; the native code overwrites it with ℓ. Six records are affected. Division-polynomial factorization and residue-field square-character checks independently account for every nonidentity torsion point.

Input candidate	ℓ	Least	Full	Relative
224 / 777	3	2	6	3
224 / 1485	3	2	6	3
224 / 777	5	2	10	5
224 / 5736	3	2	6	3
224 / 5736	13	3	39	13
256 / 876	3	2	6	3

Extension trace. Four degree-2 records incorrectly repeat the base trace. The corrected field is $t_2 = t^2 - 2p$; their factor lists already apply to that extension trace. The patched copies pass all six torsion checks and eight trace checks (degrees 1 and 2 on each curve).

```
torsion_extension.py:     k2 = l  →  k2 *= l
trace_factorization.py:   "trace": curve.trace()  →  "trace": trace
```

DiSSECT commit: 8edc3030fb91411a55aa6c07d050507d840c5e55
SageMath version 10.6, Release Date: 2025-03-31. The original default run took 169.0 seconds; internal factorization limits were retained.

FOLLOW-UP COMPUTATION

Recovered results and open factors

Three parameter results recovered

Input candidate	Trait / parameter	Result
224 / 1485	conductor, degree 4	Complete factorization via $\lvert t(t^2 - 2p)\rvert$
256 / 876	kn_factorization, k = 4	Both 4n + 1 and 4n − 1 fully factored
256 / 876	square_4p1	p = 1; order = 1: both integers are squarefree

Every recovered prime factor is proven and every integer product is checked. The degree-4 conductor-ratio identity reuses the proven base and degree-2 trace factors. The exact factors are retained in the embedded evidence.

224 / 5736 · quadratic twist over Fp²

Proven prime factors: 8581, 8548615513285513, 51858865004599249.

Unresolved cofactor (327 bits), proven composite:

```
191065098939256385908764793072579096806928265579390414297273038469196604289033778266005978480168453
```

Exact cardinality = the product of the listed prime factors and this cofactor.

256 / 876 · quadratic twist over Fp²

Proven prime factors: 1657, 45294121, 5333949541.

Unresolved cofactor (444 bits), proven composite:

```
33492266050798185653682380062483799213318454801653963493134064721059735936944520907295931496614455858076048437184499090478420204888353
```

Exact cardinality = the product of the listed prime factors and this cofactor.

Factoring scope and provenance

Distinct integer targets were deduplicated. The retry allowed 120 seconds of Sage/PARI factoring and 120 seconds of deterministic GMP-ECM per target; the exposed residuals then received a 120-second FLINT attempt each. Logs retain bounds, curve seeds and the partial factors. Residual composites have not been treated as primes.

These two open factorizations concern auxiliary groups over Fp². No statistical outlier result, runtime speedup or asymptotic attack claim follows from this report.

Primary sources and attached evidence

ICV1 specification

ICV1 reference implementation

Pinned DiSSECT source

Evidence ZIP SHA-256:

```
43c4206d682ed278aff78c4bd5b257b4a7e996edf27d7087a65a83ef04dd8e01
```

EXACT PARAMETER RECORD AND CERTIFIED IDENTITY

224-bit candidate 777

icv1-fp224-t3901553279810317031433354100437823-4b5d3f89

Supplied parameters

p	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF000000000000000000000001
a	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEFFFFFFFFFFFFFFFFFFFFFFFE
b	0x5D66353D8430A14DF0E4668AEC2D6899234A1040BAD2B4077EF9F344
Gx	0x0000000000000000000000000000000000000000000000000000000000000000
Gy	0x36BA22022A4DD98BA05F19C7D6D5594946069C1AA1F6C931FBBAB122
n	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFF3FA279701925749A7DE262AA8CC3
h	1
seed	0xD00720E7691D403449C0D91BE84BFB2C7501A267

Input values are hexadecimal except h. Derived ICV1 integers use the reference format’s decimal encoding.

Full canonical ICV1 identity

```
ICV1:fp-269599466671506397946670150870196306735579162600263081435100662988 81:3901553279810317031433354100437823
:269599466671506397946670150870196267720046364497092767101559658610 59:225824820971126244237252730250523130565 93
101247937126324318365272245:-926176686734039171949747491355849919944237301968119317895999762161 95:unk:r:4b5d3f8
9ec3f
```

Model SHA-256: `4b5d3f89ec3fd6c4325f290c8f98b68524e8f2dc0714229c05dc695054ae1945`


```
{"a":"269599466671506397946670150870196306735579162600263081435100662988 78","b":"983608904524834912214717513474 1
768612394783930987630378263670027076","field":"fp-269599466671506397946670150870196306735579162600263081435100662
298881","form":"y^2=x^3+a*x+b","p":"269599466671506397946670150870196306735579162600263081435100662988 81","v":"1
"}
```


Certified derived values

Prime full order; h = 1. Endomorphism conductor = 1. ICV1 level = unk; path = r.

Trace t = 3901553279810317031433354100437823
Embedding degree k = 44933244445251066324445025145032711286674394082848794516926609768 43 = (n − 1) / 6.

Endomorphism discriminant:

```
-926176686734039171949747491355849919944237301968119317895999762161 95
```

Base-field quadratic twist order:

```
269599466671506397946670150870196345751111960703433395768641667367 05
```

Proven prime factorization of that twist order:

```
3 × 5 × 683 × 31765017901 × 2892433723493 × 2977265004662497 × 962004386873010375632842 9
```

EXACT PARAMETER RECORD AND CERTIFIED IDENTITY

224-bit candidate 1485

icv1-fp224-t160144979725177297209579645503161-eb89557e

Supplied parameters

p	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF000000000000000000000001
a	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEFFFFFFFFFFFFFFFFFFFFFFFE
b	0x579CD953CF8611029BD2ED436A48E02C4BFE614F9A66511374CF7A8D
Gx	0x0000000000000000000000000000000000000000000000000000000000000001
Gy	0x21685513EE859C986653304C0BA16FC9E8CB143EE8ACAAB61911B9D4
n	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFF819AFA992D9C9222076DB1EF949
h	1
seed	0x18CFDBCBB825568171212BC1610A5B8A8004E5C4

Input values are hexadecimal except h. Derived ICV1 integers use the reference format’s decimal encoding.

Full canonical ICV1 identity

```
ICV1:fp-269599466671506397946670150870196306735579162600263081435100662988 81:160144979725177297209579645503161:
269599466671506397946670150870196305134129365348490109339304207957 21:241409435897007872515058163690192548926150
540603805408501543340478 32:-107814140254071381731076368453834469357966102414424367941271404203603:unk:r:eb89557
e65b2
```

Model SHA-256: `eb89557e65b2cfcfee53d31b41b321870a143f9a7343202612fef03d43050d65`


```
{"a":"269599466671506397946670150870196306735579162600263081435100662988 78","b":"922669328454351676211751026208 7
8743367707463203648113313918882597 25","field":"fp-269599466671506397946670150870196306735579162600263081435100662
298881","form":"y^2=x^3+a*x+b","p":"269599466671506397946670150870196306735579162600263081435100662988 81","v":"1
"}
```


Certified derived values

Prime full order; h = 1. Endomorphism conductor = 1. ICV1 level = unk; path = r.

Trace t = 160144979725177297209579645503161
Embedding degree k = 898664888905021326488900502900654350447097884494967031131 0140265240 = (n − 1) / 3.

Endomorphism discriminant:

```
-107814140254071381731076368453834469357966102414424367941271404203603
```

Base-field quadratic twist order:

```
269599466671506397946670150870196308337028959852036053530897118020 43
```

Proven prime factorization of that twist order:

```
3 × 31 × 10333869114542444853590048467 1 × 280526002829692975737470921944895188 1
```

EXACT PARAMETER RECORD AND CERTIFIED IDENTITY

224-bit candidate 5736

icv1-fp224-t32245555999540959732222272472665529-b7920759

Supplied parameters

p	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF000000000000000000000001
a	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEFFFFFFFFFFFFFFFFFFFFFFFE
b	0x4D41BDD4F0ADA5E7746A76329E4FD5DD879E04ADBFE7FCD2F5756748
Gx	0x0000000000000000000000000000000000000000000000000000000001
Gy	0x7AFEFAB438B7A9EE10F354AE4A4DE44388B0EED987AC09D625877E75
n	0xFFFFFFFFFFFFFFFFFFFFFFFFFFFF610362EB9E61800D49507468B249
h	1
seed	0x3C170A30EE6709DF9B4C9EA0B6C1E95FCD6200CE

Input values are hexadecimal except h. Derived ICV1 integers use the reference format’s decimal encoding.

Full canonical ICV1 identity

```
ICV1:fp-269599466671506397946670150870196306735579162600263081435100662988
81:32245555999540959732222724726655
29:269599466671506397946670150870196274490023163059303349212375936333
53:10806199813798849062791590979138478661110
430734462130877561845822529:-9744202785140723935188329567548515544053065039837095023756196034568
3:unk:r:b792075
970b9
```

Model SHA-256: `b792075970b9a6906b3d6799ac1ca495c3ecb576b7245d7a2caaeb2e95f1414d`


```
{"a":"269599466671506397946670150870196306735579162600263081435100662988
78","b":"81360909554727208821243875699177821513153859023339441914346726623
44","field":"fp-269599466671506397946670150870196306735579162600263081435100662
98881","form":"y^2=x^3+a*x+b","p":"269599466671506397946670150870196306735579162600263081435100662988
81","v":"1
"}
```


Certified derived values

Prime full order; h = 1. Endomorphism conductor = 1. ICV1 level = unk; path = r.

Trace t = 32245555999540959732222724726655 29
Embedding degree k = 898664888905021326488900502900654248300077210197677830707919787778 4 = (n − 1) / 3.

Endomorphism discriminant:

```
-9744202785140723935188329567548515544053065039837095023756196034568
3
```

Base-field quadratic twist order:

```
269599466671506397946670150870196338981135162141222813657825389644
11
```

Proven prime factorization of that twist order:

```
3 × 1607 × 3371 × 17539 × 20347 × 1231313 × 71562375091153 × 527551328299791166478629025047
33
```

EXACT PARAMETER RECORD AND CERTIFIED IDENTITY

256-bit candidate 876

icv1-fp256-t1058672357239064776925535251936763850 09-a67f0bc0

Supplied parameters

p	0xFFFFFFFF000000010000000000000000000000FFFFFFFFFFFFFFFFFFFFFFFF
a	0xFFFFFFFF000000010000000000000000000000FFFFFFFFFFFFFFFFFFFFFFFC
b	0xA84FCA4BE639727675CCEA4A9CECDA4B900C32DD7E86192AB71B355FDE8255BD
Gx	0x0000000000000000000000000000000000000000000000000000000000000000
Gy	0x0E97F569CECFCE57BCAC0976EB9228FFA9FD271940DB013407840C8DD17B3B7A
n	0xFFFFFFFF00000000FFFFFFFFFFFFFFFFB05AB64A7F5B43FA0A2D70FED440950F
h	1
seed	0x50C71A63F911DB7B2F31618ECF609D37154437F9

Input values are hexadecimal except h. Derived ICV1 integers use the reference format’s decimal encoding.

Full canonical ICV1 identity


```
ICV1:fp-1157920892103562487626974469494075735300861434152903141955336313088670978539 51:1058672357239064776925535251936763850 09:1157920892103562487626974469494075735299802761795664077178410777836734214689 43:1666431558791302003980351035011959112391866348083660076214606213177407344151 1:-45196048524160381514294955574887608279554759907305149886414376967531399148 5723:unk:r:a67f0bc0376d
```


Model SHA-256: `a67f0bc0376d0032c0b5edcc05d775b92b385bb9234dadff8f82363fbc1edace`


```
{"a":"1157920892103562487626974469494075735300861434152903141955336313088670978539 48","b":"761295356791097567971720748918501105824622924068619648339139894146183820958 05","field":"fp-1157920892103562487626974469494075735300861434152903141955336313088 67097853951","form":"y^2=x^3+a*x+b","p":"1157920892103562487626974469494075735300861434152903141955336313088670978539 51","v":"1"}
```


Certified derived values

Prime full order; h = 1. Endomorphism conductor = 1. ICV1 level = unk; path = r.

Trace t = 1058672357239064776925535251936763850 09
Embedding degree k = 3859736307011874958756581564980252450999342539318880257261369259455780715631 4 = (n − 1) / 3.

Endomorphism discriminant:

```
-45196048524160381514294955574887608279554759907305149886414376967531399148 5723
```

Base-field quadratic twist order:

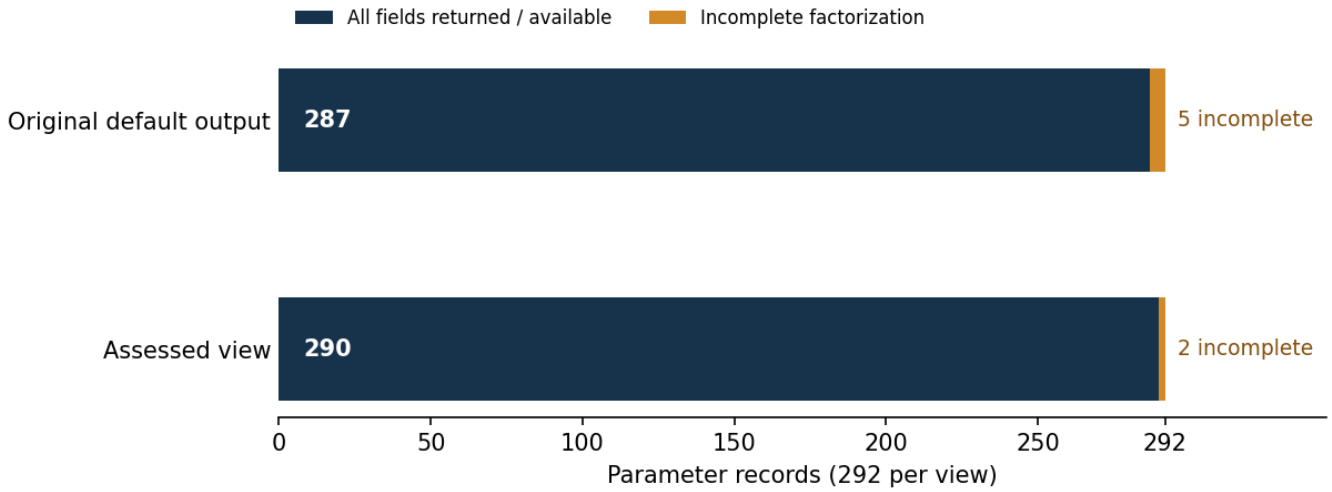
```
1157920892103562487626974469494075735301920106510142206732261848340607742389 61
```

Proven prime factorization of that twist order:

```
3 × 11 × 37 × 8081 × 5912232803 × 19849363840034333991855225188348685618817088905528103546614 87
```

Result coverage and provenance

The upstream default run and the assessed view each contain 292 parameter records. The graph shows field availability only. Ten original records contain the verified source defects described on page 2; numerical availability alone does not certify a result.



Source: continuation/continuation-summary.json; editable plotting source and graph-source.json accompany the report. The counts are deterministic accounting for this single four-curve run; no uncertainty interval or sampled population is implied.

How to read the evidence

Layer	Contents	Role
Supplied inputs	Four parameter sets, seeds and generator conventions	Exact requested models
Frozen DiSSECT output	22 traits, 292 default calls	Original observations and timeouts
Independent checks	Order, factors, embedding degrees and point coordinates	Evidence for stated arithmetic conclusions
Assessed view	Ten corrected records; three recovered parameter results	Versioned interpretation with field-level provenance
Open obligations	Two composite cofactors over Fp^2	Incomplete factorization; no impossibility claim

No IC/rho timing comparison, solver implementation, global curve-registry update or leaderboard promotion is made by archiving this report.